

Question 1:

Question: A company is considering two financing options for a new project:

- **Option A:** Issue \$2,000,000 in bonds with a 5% annual coupon rate, paid semi-annually, and a 10-year maturity.
- **Option B:** Issue 100,000 shares of common stock at \$20 per share, with an expected annual dividend of \$1.50 per share.

- a. For **Option A**, calculate the total interest payment the company will make in the first year.
- b. For **Option B**, if the company's earnings are projected to be \$800,000 annually, what percentage of earnings will be paid out as dividends in the first year?
- c. Compare the two options in terms of fixed obligations (interest vs. dividends) and explain which one may be preferable if the company expects volatile cash flows.

Type: short-answer

Options: No options provided.

Question 2:

Question: In a sole proprietorship, how are business profits typically taxed, and what is the owner's personal liability for business debts?

Type: multiple-choice

Options:

- A. Profits are taxed at the corporate tax rate; owner has unlimited personal liability
- B. Profits are taxed as personal income; owner has unlimited personal liability
- C. Profits are taxed as personal income; owner has limited liability to their investment
- D. Profits are taxed at the corporate tax rate; owner has limited liability to their investment

E. Profits are exempt from taxation; owner has no personal liability

****Knowledge Points:**** Taxation of sole proprietorships, Liability in sole proprietorships

Question 3:

****Question:****

A bond with a face value of \$1,000 has a coupon rate of 4% and is currently priced at \$950.

The bond matures in 5 years. Analyze the following statements and determine which are correct:

- I. The bond is trading at a discount to its face value.
- II. The yield to maturity (YTM) is greater than the coupon rate.
- III. If market interest rates rise, the bond's price will increase as it approaches maturity.
- IV. The bond's price sensitivity to interest rate changes is higher than that of a similar bond with a 6% coupon rate.

****Type:**** multiple-choice

****Options:****

- A. I and II only
- B. I, II, and IV only
- C. II and III only
- D. I, III, and IV only
- E. I, II, III, and IV

****Knowledge points:****

- Discount bond price relative to face value
- Relationship between YTM and coupon rate
- Bond price behavior as maturity approaches
- Coupon rate impact on interest rate sensitivity

Question 4:

Question:

Consider two bonds issued by the same company with the same credit rating and yield to maturity (YTM). Bond A is a 10-year zero-coupon bond, while Bond B is a 10-year bond with a 5% annual coupon rate. If market interest rates increase by 1%, which bond will experience a greater percentage decline in price? Explain the factors that contribute to this difference in price sensitivity.

Type: short-answer

Knowledge Points:

- Zero-coupon bond price sensitivity to interest rate changes
- Impact of coupon payments on bond duration and interest rate risk
- Relationship between bond duration and price volatility

Question 5:

Question: A bank offers two different savings accounts. Account X has an APR of 5% compounded monthly, while Account Y has an APR of 5.1% compounded annually. If you plan to deposit \$10,000 and leave it untouched for 5 years, which account should you choose to maximize your returns?

****Options:****

- A. Account X; the effective annual rate is higher.
- B. Account Y; the annual percentage rate is higher.
- C. Account X; the period rate is higher.
- D. Account Y; the effective annual rate is higher.
- E. Both accounts will yield the same return after 5 years.

Question 6:

****Question:****

ABC Corporation's common stock is currently priced at \$50 per share. The company just paid a dividend of \$2.00 per share, and dividends are expected to grow at a constant rate of 5% per year indefinitely. What is the dividend yield and capital gains yield for ABC Corporation's stock?

****Type:**** short-answer

****Options:**** No options provided.

****Knowledge points:**** Dividend yield, Capital gains yield, Constant growth stock (Gordon Growth Model)

Question 7:

****Question:****

A corporate bond with a face value of \$1,000 pays semi-annual coupons and is currently priced at \$1,150. The bond has a yield to maturity (YTM) of 5.5% and matures in 10 years. What is the bond's annual coupon rate?

****Type:**** multiple-choice

****Options:****

A. 4.50%

B. 5.00%

C. 6.00%

D. 6.50%

E. 7.00%

****Knowledge points assessed:****

- Present value formula for semi-annual bonds
- Coupon rate calculation from bond price
- Relationship between bond price, YTM, and coupon payments

Question 8:

****New Question:****

Pinnacle Enterprises recently paid a dividend of \$5 per share (D_0). The company expects dividends to grow at a high rate of 12% annually for the next six years. After this high-growth phase, the dividend growth rate will stabilize at 5% per year indefinitely. The required rate of return for Pinnacle's stock is 9%.

- Calculate the present value of the dividends during the high-growth phase (Years 1–6).
- Determine the terminal value of the stock at the end of the high-growth phase (Year 6).
- Compute the intrinsic value of Pinnacle's stock today by discounting the terminal value and summing it with the present value of the high-growth dividends.

****Explain how changes in the required return (e.g., an increase to 10%) would affect each component of your valuation (high-growth phase, terminal value, and total stock price).****

Type: short-answer

Options: No options provided.

Knowledge Points:

- Two-stage Dividend Discount Model (high growth + stable growth)
- Present value calculation of growing annuity (high-growth phase)
- Terminal value estimation using Gordon Growth Model
- Sensitivity of stock valuation to changes in discount rate

Question 9:

Question:

You are considering two investment options:

1. **Annuity A** pays \$3,000 at the **beginning** of each year for 15 years.
2. **Annuity B** pays a fixed amount at the **end** of each year for 15 years.

Both annuities have the same present value when discounted at an annual rate of 7.5%.

What is the annual payment of Annuity B?

Type: multiple-choice

Options:

A. \$2,850

B. \$3,000

C. \$3,225

D. \$3,450

E. \$3,600

(Knowledge points: Annuity present value calculations, Ordinary annuities vs. annuities due, Solving for payment in financial contexts)

Question 10:

****Question:****

Consider the following statements about the term structure of interest rates:

I. An upward-sloping yield curve implies that market participants expect higher inflation in the future.

II. A flat yield curve suggests that real interest rates are expected to decline over time.

III. A downward-sloping (inverted) yield curve may indicate that investors are demanding a higher interest rate risk premium for long-term bonds.

IV. The term structure reflects both inflation expectations and the real interest rate, but not liquidity preferences.

Which of the statements above are correct?

****Type:**** multiple-choice

****Options:****

A. I and II only

B. I and III only

C. II and III only

D. I, II, and III only

E. I, III, and IV only

(Analyze the components of the term structure, differentiate between inflation expectations, real rates, and risk premiums, and evaluate the validity of each statement.)

Question 11:

Question:

You are planning for retirement and want to determine how much you need to save annually to reach your goal. You currently have \$50,000 in your retirement account and aim to accumulate \$1,500,000 by the time you retire in 25 years. The account earns an annual interest rate of 6%, compounded monthly.

(a) What is the effective annual rate (EAR) for this account?

(b) How much must you deposit at the **end** of each year to reach your retirement goal, given your current savings?

(c) If you instead make deposits at the **beginning** of each year, how much would your annual deposit need to be?

Type: long-answer

Options: No options provided.

(Assesses: Effective annual rate calculation, future value of annuity for retirement planning, ordinary annuity vs. annuity due adjustments.)

Question 12:

Question:

A bond with a face value of \$1,000 pays a 6% annual coupon and is currently priced at \$1,120. The bond has 8 years remaining until maturity.

- a) Calculate the current yield of the bond.
- b) If the yield to maturity (YTM) decreases by 1.5% in two years and you sell the bond at that time, what will be the selling price?
- c) What is the holding period return (HPR) if you buy the bond today and sell it after two years under the conditions in part (b)?

Type: short-answer

Options: No options provided.

Knowledge Points:

- Current yield calculation for bonds
- Bond pricing with changing YTM
- Holding period return (HPR) determination

Question 13:

Question:

You are considering taking out a \$300,000 mortgage to buy a house. The loan has a 30-year term with a fixed annual interest rate of 4.5%, compounded monthly.

- a. Calculate the **monthly mortgage payment** required to fully amortize the loan over the 30-year period.
- b. After making payments for 10 years, you decide to refinance the remaining balance at a

new rate of 3.75% for the remaining 20 years. What will be your ****new monthly payment**** after refinancing?

c. How much ****total interest**** will you save over the life of the loan by refinancing after 10 years?

****Type:**** short-answer

****Options:**** No options provided.

****Knowledge Points:**** Loan amortization, Present value of an annuity, Time value of money, Refinancing calculations.

Question 14:

****Question:****

TechGrow Inc. just paid a dividend of \$5 per share. The company expects its dividends to grow at a rapid rate of 18% annually for the next five years. After this period, the growth rate will stabilize at a constant rate of 5% per year indefinitely. If the required rate of return for TechGrow's stock is 12%, what is the current value of the stock today?

****Type:**** short-answer

****Options:**** No options provided.

****Knowledge Points:**** Dividend Discount Model (DDM) two-stage growth model, Present value of high-growth phase dividends, Terminal value calculation post high-growth phase.